Python Engineering - Class 02 Reference Guide

Conditionals and Special Cases

Reference guide to the syntax and tools used in this class. It explains what each element does, how it is written and when it is useful.

Quick map

Concept	Syntax	Idea
Boolean	True / False	Result of comparisons and logical expressions.
if	if condition:	Executes a block only when the condition is True.
elif	elif condition:	Checks another condition only if previous branches were False.
else	else:	Fallback branch when no previous condition is True.
Comparisons	== != < <= > >=	Create Boolean expressions.
Logical operators	and / or / not	Combine or negate Boolean expressions.
isinstance()	isinstance(x, numbers.Real)	Check whether an object belongs to a type/class family.
np.log()	np.log(x)	Natural logarithm, used by the rocket equation.
np.sqrt()	np.sqrt(x)	Square root; domain and complex cases must be considered.
sys.exit()	sys.exit()	Stops script execution; later classes replace some uses with functions/exceptions.

1. Boolean

Syntax: `True / False`

Result of comparisons and logical expressions.

```
is_valid = m0 > mf
```

Control question: before running the example, predict the result type and shape.

2. if

Syntax: `if condition:`

Executes a block only when the condition is True.

```
if speed < 0:
    print('Invalid speed')
```

Control question: before running the example, predict the result type and shape.

3. elif

Syntax: `elif condition:`

Checks another condition only if previous branches were False.

```
if M < 0.8:
    Cd = 0.3
elif M < 1.2:
    Cd = 0.5
```

Control question: before running the example, predict the result type and shape.

4. else

Syntax: `else:`

Fallback branch when no previous condition is True.

```
else:
    Cd = 0.25
```

Control question: before running the example, predict the result type and shape.

5. Comparisons

Syntax: `== != < <= > >=`

Create Boolean expressions.

```
D = b**2 - 4*a*c
if D < 0:
    ...
```

Control question: before running the example, predict the result type and shape.

6. Logical operators

Syntax: `and / or / not`

Combine or negate Boolean expressions.

```
if m0 > 0 and mf > 0:
    ...
```

Control question: before running the example, predict the result type and shape.

7. isinstance()

Syntax: `isinstance(x, numbers.Real)`

Check whether an object belongs to a type/class family.

```
if not isinstance(N, int):
    ...
```

Control question: before running the example, predict the result type and shape.

8. np.log()

Syntax: `np.log(x)`

Natural logarithm, used by the rocket equation.

```
delta_v = Isp * g0 * np.log(m0 / mf)
```

Control question: before running the example, predict the result type and shape.

9. np.sqrt()

Syntax: `np.sqrt(x)`

Square root; domain and complex cases must be considered.

```
root = np.sqrt(value)
```

Control question: before running the example, predict the result type and shape.

10. sys.exit()

Syntax: `sys.exit()`

Stops script execution; later classes replace some uses with functions/exceptions.

```
if T <= -273:
    sys.exit()
```

Control question: before running the example, predict the result type and shape.

Common mistakes

- Confusing a function name with a function call using parentheses.
- Not checking types, indices or shapes before an operation.
- Using a tool because it works without being able to explain what it receives and returns.

Mental cheat sheet

```
Boolean: True / False
if: if condition:
elif: elif condition:
else: else:
Comparisons: == != < <= > >=
Logical operators: and / or / not
isinstance(): isinstance(x, numbers.Real)
np.log(): np.log(x)
np.sqrt(): np.sqrt(x)
sys.exit(): sys.exit()
```